STA2002: Probability and Statistics II
Order Statistics

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Introduction

- In this lecture we will introduce the concept of **order statistics** before we can proceed to talk about QQ plots or distribution-free hypothesis testing.

- **Suggested reading**: Chapter 6.3 of the textbook.

Order Statistics

can be used in distribution free (hypotheses testing)

a function of sample $x_1, x_2 \cdots x_n$

Definition 1 (order statistics)

Suppose that $X_1, \ldots, X_n$ are i.i.d. random samples from a common distribution. For any $k = 1, 2, \ldots, n$, the kth order statistic is the random variable

$$X_{(k)} = \text{kth smallest of } X_1, \ldots, X_n.$$

For example,

$$\begin{aligned} X_{(1)} &= \min\{X_1, X_2, \ldots, X_n\}, \\ X_{(2)} &= \text{second smallest of } X_1, \ldots, X_n, \\ X_{(n)} &= \max\{X_1, X_2, \ldots, X_n\}. \end{aligned}$$

Random sample $x_1 \cdots x_n$ iid X

Ordered sample $X_{(1)} \leq X_{(2)} \leq \cdots \leq X_{(n)}$.

Question: The distributions of X_1 and $X_{(i)}$

Practice: To derive the distribution of some order statistics

Example: Consider random sample $X_1 \cdots X_5$

Task: ① To derive the cdf of $X_{(5)}$

② To derive the pdf of $X_{(5)}$

Example 6.3-1 and 6.3-2

Example 2

Let $X_1, \ldots, X_5$ be i.i.d. random samples from the pdf

$$f(x) = 2x, \quad 0 < x < 1.$$

Its cdf is, for $0 < x < 1$,

$P(X_i \le x) = P(X \le x)$
$= F_X(x)$

$$F(x) = \int_0^x 2y\, dy = x^2.$$

The values $x_1 = 0.62, x_2 = 0.98, x_3 = 0.31, x_4 = 0.81$, and $x_5 = 0.53$ are the observed values.

Example 6.3-1 and 6.3-2

- The observed order statistics are

$$
\begin{aligned}
x_{(1)} &= \min\{x_1, x_2, x_3, x_4, x_5\} = 0.31 \\
x_{(2)} &= 0.53 \\
x_{(3)} &= 0.62 \\
x_{(4)} &= 0.81 \\
x_{(5)} &= \max\{x_1, x_2, x_3, x_4, x_5\} = 0.98
\end{aligned}
$$

- Question: What is the cdf of $X_{(5)}$?

Example 6.3-1 and 6.3-2

- Consider the cdf of $X_{(5)}$:

$$
\begin{aligned}
P\left(X_{(5)} \leqslant x\right) &= P\left(\max\{X_1, X_2, X_3, X_4, X_5\} \leqslant x\right) \\
&= P\left(X_1 \leq x, X_2 \leq x, \ldots, X_5 \leq x\right) \\
&= \prod_{i=1}^{5} P\left(X_1 \leq x\right) \qquad \text{by i.i.d.} \\
&= x^{10} = F(x)^5.
\end{aligned}
$$

- We differentiate the above cdf to obtain the pdf of $X_{(5)}$, that we denote by $f_{X_{(5)}}$:

$$
f_{X_{(5)}}(x) = 10x^9 = 5F(x)^4 f(x), \quad 0 < x < 1.
$$

Example 6.3-1 and 6.3-2

- Question: What is the cdf of $X_{(4)}$?
- Define a random variable W, where W is the number of X_i such that $X_i \leq x$.
- Since $X_1, \ldots, X_5$ are i.i.d., $W \sim Bin(5, F(x))$, i.e. the "success probability" is $P(X_i \leq x) = F(x)$.

Construction of W.
frequence of success.
random trial.
Binomial → {success $X_i \leq x$, failure $X_i > x$}

$$
\begin{aligned}
P\left(X_{(4)} \leqslant x\right) &= P\left(\text{at least 4 of } X_1, \ldots, X_5 \leq x\right) \\
&= P\left(W \geq 4\right) \\
&= P(W = 4) + P(W = 5) \\
&= \binom{5}{4} F(x)^4 (1 - F(x))^1 + F(x)^5.
\end{aligned}
$$

try to transform event into equivalent event.

Example 6.3-1 and 6.3-2

- Differentiate the above cdf w.r.t. x to obtain the pdf of $X_{(4)}$, which is denoted as $f_{X_{(4)}}$:

$$
\begin{aligned}
f_{X_{(4)}}(x) &= \binom{5}{4} 4F(x)^3 f(x)(1-F(x)) \\
&\quad + \binom{5}{4} F(x)^4(-f(x)) + 5F(x)^4 f(x) \\
&= \binom{5}{4} 4F(x)^3 f(x)(1-F(x)) \\
&= \frac{5!}{3!1!1!} F(x)^3 f(x)(1-F(x)), \quad 0 < x < 1.
\end{aligned}
$$

cdf and pdf Formula for Order Statistics

- We now generalize the results in the following theorem.

Theorem 3

Suppose that $X_1, \ldots, X_n$ are i.i.d. continuous random variables with common pdf $f(x)$ and cdf $F(x)$. For $k = 1, \ldots, n$, denote the cdf and pdf of the kth order statistic $X_{(k)}$ to be respectively $F_{X_{(k)}}$ and $f_{X_{(k)}}$. They can be written as

$$F_{X_{(k)}}(x) = \sum_{l=k}^{n} \binom{n}{l} F(x)^l (1 - F(x))^{n-l},$$

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!1!(n-k)!} [F(x)]^{k-1} f(x) [1 - F(x)]^{n-k}.$$

cdf and pdf Formula for Order Statistics

The interpretation is the following:

- $F(x)^{k-1}$ corresponds to the probability that there are $k-1$ samples less than x
- $f(x)$ corresponds to the probability that there are 1 sample around x, namely $X_{(k)}$
- $(1-F(x))^{n-k}$ corresponds to the probability that there are $n-k$ samples greater than x
- The multinomial coefficient $\frac{n!}{(k-1)!\ 1!\ (n-k)!}$ in the front captures all the different combinations in the ordering of the sample items

Proof of Theorem

Proof.

- We define $W \sim Bin(n, F(x))$. That is, W is the number of X_i such that $X_i \leq x$.
- Then cdf of $X_{(k)}$ is,

$$\begin{aligned} F_{X_{(k)}}(x) &= P(X_{(k)} \leq x) \\ &= P(W \geq k) \\ &= \sum_{l=k}^{n} P(W = l) \\ &= \sum_{l=k}^{n} \binom{n}{l} F(x)^l (1 - F(x))^{n-l} \end{aligned}$$

Proof of Theorem (Cont'd)

- Differentiate cdf w.r.t. x to obtain the pdf,

$$
\begin{aligned}
f_{X_{(k)}}(x) &= \sum_{l=k}^{n-1} \binom{n}{l} l[F(x)]^{l-1} f(x)[1-F(x)]^{n-l} + \sum_{l=k}^{n-1} \binom{n}{l} (n-l)[F(x)]^{l}(-f(x))[1-F(x)]^{n-l-1} \\
&\quad + nF(x)^{n-1} f(x) \\
&= \sum_{l=k}^{n} \binom{n}{l} l[F(x)]^{l-1} f(x)[1-F(x)]^{n-l} + \sum_{l=k}^{n-1} \binom{n}{l} (n-l)[F(x)]^{l}(-f(x))[1-F(x)]^{n-l-1} \\
&= \sum_{l=k}^{n} \binom{n}{l} l[F(x)]^{l-1} f(x)[1-F(x)]^{n-l} \\
&\quad + \sum_{l'=k+1}^{n} \binom{n}{l'-1} (n-l'+1)[F(x)]^{l'-1}(-f(x))[1-F(x)]^{n-l'} \\
&= \sum_{l=k}^{n} \frac{n!}{(l-1)!(n-l)!}[F(x)]^{l-1} f(x)[1-F(x)]^{n-l} \\
&\quad + \sum_{l'=k+1}^{n} \frac{n!}{(l'-1)!(n-l')!}[F(x)]^{l'-1}(-f(x))[1-F(x)]^{n-l'} \\
&= \frac{n!}{(k-1)!1!(n-k)!}[F(x)]^{k-1} f(x)[1-F(x)]^{n-k}.
\end{aligned}
$$

□

Order Statistics of Uniform Distribution

- Suppose that $U_1, \ldots, U_n \overset{i.i.d.}{\sim} Uniform(0,1)$. Then its cdf and pdf are

$$F_{U_1}(x) = x, \quad f_{U_1}(x) = 1 \quad 0 < x < 1.$$

- Denote the order statistics by $U_{(1)}, U_{(2)}, \ldots, U_{(n)}$.
- By Theorem 3, the pdf of $U_{(k)}$ for $k = 1, \ldots, n$ is

$$\begin{aligned} f_{U_{(k)}}(x) &= \frac{n!}{(k-1)!1!(n-k)!} x^{k-1}(1-x)^{n-k} \\ &= \frac{n!}{(k-1)!(n-k)!} x^{k-1}(1-x)^{n-k+1-1} \\ &= \frac{\Gamma(k+n-k+1)}{\Gamma(k)\Gamma(n-k+1)} x^{k-1}(1-x)^{n-k+1-1}. \end{aligned}$$

Order Statistics of Uniform Distribution (Cont'd)

- **Remark**: The distribution of $U_{(k)}$ belongs to beta distribution,

$$U_{(k)} \sim Beta(k, n-k+1).$$

The mean is,

$$E(U_{(k)}) = \frac{k}{n+1}.$$

Definition 4 (Beta Distribution)

A random variable X is said to follow the **Beta distribution** with parameters $\alpha, \beta > 0$, written as $X \sim Beta(\alpha, \beta)$, if its pdf can be written as

$$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}, \quad 0 < x < 1.$$

Mean and Variance of Beta Distribution

- The mean and variance of X which follows a beta distribution with parameters α and β are,

$$E(X) = \frac{\alpha}{\alpha + \beta},$$

and

$$Var(X) = \frac{\alpha\beta}{(\alpha + \beta + 1)(\alpha + \beta)^2}.$$

Distributions of $F(X_k)$

- Suppose that $X_1, \ldots, X_k$ are i.i.d. continuous random variables with strictly increasing cdf F.
- We would show that $F(X_k) \sim Uniform(0, 1)$.
- Define $Y = F(X_k)$, then its cdf is, for $0 < y < 1$,

$$\begin{aligned} P(Y \leq y) &= P(F(X_k) \leq y) \\ &= P(X_k \leq F^{-1}(y)) \\ &= F(F^{-1}(y)) \\ &= y, \end{aligned}$$

which is exactly the cdf of Uniform(0,1).

Distributions of $F(X_{(k)})$

- Suppose that $X_1, \ldots, X_k$ are i.i.d. continuous random variables with strictly increasing cdf F.
- We would show that $F(X_{(k)}) \sim Beta(k, n-k+1)$, where $X_{(k)}$ denotes the kth order statistic.
- We have the following mathematical deduction,

$$\begin{aligned} F(X_{(k)}) &= F \text{ evaluated at } X_{(k)} \\ &= \text{ kth smallest of } F(X_1), F(X_2), \ldots, F(X_n) \\ &\quad \text{since } X_i < X_j \Leftrightarrow F(X_i) < F(X_j) \\ &= \text{ kth smallest of } U_1, U_2, \ldots, U_n \\ &= U_{(k)} \sim Beta(k, n-k+1). \end{aligned}$$